

RESEARCH ARTICLE

Improving Water-Based Ink Adhesion on Commercial Polymers by Corona Treatment

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ABSTRACT

Polyolefins are widely employed as external skins for flexible packaging, yet their nonpolar, hydrophobic surfaces present challenges for high-resolution, wear-resistant printing. Converters have traditionally relied on surface treatments, such as corona treatment, to increase surface free energy (SFE) and promote compatibility with solvent-based inks. While water-based inks offer sustainability advantages and reduced volatile organic compound exposure over solvent-based inks, their adhesion to polymer substrates is inconsistent and generally not robust. Industry guidelines suggest SFE differentials between ink and substrate are the only factor influencing adhesion; however, this factor may not completely describe the ink-substrate interactions. The adhesion performance of water-based inks was evaluated on polyethylene, ethylene acrylic acid copolymer, ethylene vinyl alcohol copolymer, and polyamide. Polymer films were examined pre- and post-corona treatment to assess changes in surface energy, chemistry, and topography on ink printability and durability. To achieve acceptable print durability, these polymers all required corona treatment, leading to chemical and morphological modifications to the film surface. Results demonstrated that surface chemistry and topography are stronger indicators of water-based ink adhesion than SFE alone. These findings establish a framework for optimizing the durability of water-based inks on polyolefins and related substrates through targeted surface treatment strategies.

1 | Introduction

The surface printing of flexible polymer packaging is crucial for aesthetic and informational reasons. Enhancing a product's attractiveness and functionality to consumers is a significant factor in the supply chain processes involved in finished goods production. Flexible packaging articles are typically produced with polyolefins (e.g., polyethylene) as external skin layers due to their abuse resistance, durability, flexibility, and corrosion resistance properties, in addition to low cost

and ease of processing. However, due to polyolefins' chemical structure and their resulting low surface free energy (SFE), these polymers cannot form strong bonds with ink, overprint varnish, or lacquer coatings [1]. To overcome adhesion issues common with low SFE materials, packaging producers (referred to in the industry as converters) apply a combination of surface treatment and organic solvent-based ink systems to enable surface printing of these materials. With the global movement towards more sustainable solutions, traditional organic solvent-based ink systems have been criticized due to

Summary

- Polymer films were printed with water-based ink to evaluate print durability.
- High native surface free energy polymers did not display higher print durability.
- Polymers tested required corona treatment to have acceptable durability.
- Surface chemistry and topography determine the success of ink adhesion.

the environmental impacts of the raw materials of the inks [2]. Additionally, there are health concerns for industry workers over prolonged exposure to these inks associated with inhalation and contact with volatile organic compounds present in organic solvents [3]. Water-based ink systems are considered more environmentally friendly, eliminate workers' exposure to organic solvents, and can produce higher-definition imaging than solvent-based inks [4]. Digital printing, also called inkjet printing, which uses water-based ink systems, has been touted as the future of printing technology for packaging graphics [5, 6]. This research examines how polymer selection and corona surface treatment impact the printability of water-based ink systems.

Interactions between inks and substrates are critical during printing. The adhesion of the ink to the outer layer of the packaging film is hypothesized to be determined by the interactions between polar and nonpolar (dispersive) components of the inks and the polymer films. The greater the similarity between the polar and dispersive components of inks and substrates, the better the wetting and adhesion on the surface of the printing substrate. Increasing the surface density of polar functional groups and the resulting increase in SFE can overcome poor wetting and adhesion on polymer surfaces. A guideline commonly used in industry is that the SFE of the printing substrate typically needs to be 15–20 dynes/cm higher than the ink's surface tension to achieve proper adhesive bonding. To the authors' knowledge, no comprehensive study has critically assessed the overarching applicability and limitations of this industrial guideline. This study evaluates this widely used industry guideline on surface energy differences, specifically studying SFE and other physical and chemical factors that may contribute to ink adhesion and durability.

There are several methods that converters can use to modify polymer film surfaces prior to printing, promoting ink adhesion, including corona treatment, flame treatment, plasma treatment, and chemical treatment [7, 8]. Generally, these surface treatments are designed to increase the SFE of the film by ~15–20 dynes/cm compared to the neat film by introducing reactive groups that bond with the ink on the surface [9–13]. A typical water-based ink has a surface tension in the range of 25–30 dynes/cm. Using the industry guideline that the substrate surface tension should be 15–20 dynes/cm higher than the ink, the target surface energy of the print substrate should be 40–50 dynes/cm to achieve acceptable adhesion [3].

Methods such as corona treatment are widely used in industry to promote surface oxidation that can increase the adhesion between solvent-based inks and polyolefin surfaces [9]. When a film is subjected to corona treatment, the treatment produces polar and hydrophilic functional groups on the surface, such as hydroxyl ($-\text{OH}$), carboxyl ($\text{R}-\text{COOH}$), carbonyl ($\text{C}=\text{O}$), or ester ($\text{R}-\text{CO}-\text{OR}$) groups [10, 13]. Introducing these polar groups on the film surface increases the SFE and subsequently improves the wettability and adhesion properties between the liquid ink and the polymer film [13]. A drawback of corona treatment is that the treatment effects decrease over time as non-functionalized polymer chains diffuse from the bulk to the surface. As a result, films often need to be retreated if they are not printed immediately [14, 15]. While these surface treatments are commonly used in typical packaging polyolefins, they are not always available (capital expense), may not produce adequate oxidation to increase the surface energy of these films and have the limitation of needing to print very soon after treatment. Another consideration is that different polymers have varied responses to these surface treatments, resulting in variable levels of oxidation and, in some cases, detrimental responses to treatment, such as chain scission [10].

A wide variety of polymers are used in packaging films, each exhibiting a range of surface and bulk properties. This study focuses on a group of polymers extensively utilized in packaging films to explore how their surface characteristics influence printability and print stability. Polyethylene is used as a model nonpolar polymer for comparison with the other selected polymers. The selected polymers in this study that possess inherent latent functionality include ethylene acrylic acid copolymer (EAA), ethylene vinyl alcohol block copolymer (EVOH), and polyamide (PA) (Figure 1).

Polyethylene (PE) is the most commonly used polymer in flexible packaging [9]. PE has the least complex structure in polymers, with only methylene (CH_2) repeat units, which are nonpolar and relatively inert. PE and other nonpolar polymer films are difficult to print on due to their nonpolar and hydrophobic nature, as indicated above. Without any hydrophilic functional groups, PE must be surface-treated to create functional groups that can provide favorable interactions with the inks to improve printing.

EAA copolymers are used in the packaging industry to adhere to polar substrates or act as interlayer adhesion layers between coextruded multilayer polymer films [16]. Occasionally, EAA has also been used on external surfaces where increased adhesion is desired for the surface. The carboxylic acid groups from the acrylic acid content [10] in EAA can serve as reactive functional groups and orient to the surface, and this polar surface functionality promotes adhesion with polar inks.

EVOH was selected as one of the polymers due to its surface energy value of 41–45 dynes/cm [17], which is comparable to corona-treated PE. EVOH is a random copolymer of PE and poly(vinyl alcohol) (PVA). PVA contains an aliphatic $\text{C}-\text{C}$ backbone and pendant $-\text{OH}$ groups, and these polar groups render the copolymer hydrophilic [17]. EVOH is commonly used in packaging as a barrier layer to protect the contents due to its performance as a barrier to various gases. The selection

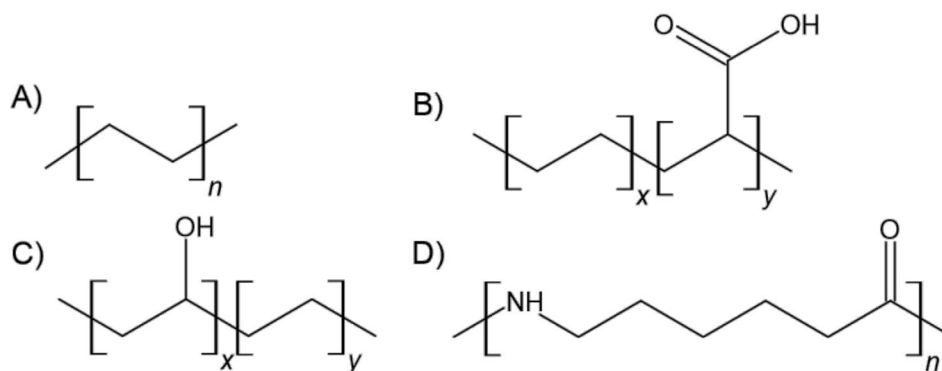


FIGURE 1 | Chemical structures of polymers considered here: PE (A), EAA (B), EVOH (C), and PA (D).

of EVOH for this study is based on its hydrophilic nature, desirable SFE values, and functional groups available for interactions with ink.

PA also has a higher native surface energy that falls in the typical range of 48–51 dynes/cm [18] and has available amide functional groups ($-CONH_2$). Commercial polyamide comes in a variety of grades based on monomer structure (e.g., PA6, PA66, copolymer PA6/66). PA6/66 is often used in packaging for film strength, barrier properties, and improved processing compared to PA6 and PA66 grades, and can be used throughout multilayer structures, including as an exterior skin layer. Due to its prevalent use in commercial applications, the copolymer PA6/66 is the form of PA selected for this study.

Most of the literature data targeting improved adhesion between water-based ink and polymer film surfaces has focused on the ink composition [19, 20]. Limited research has been performed that modifies the polymers and their surface characteristics while maintaining the constant ink composition. Also, how the surface treatment alters the surface topography and correspondingly changes the ink adhesion behavior is not well understood. This study aims to provide a new understanding of the critical surface characteristics of polymer films to achieve acceptable ink adhesion and durability in digital print applications.

2 | Experimental

2.1 | Materials

2.1.1 | Film Preparation

The test samples used in this study are multilayer polymer films produced on a commercial coextrusion line at Sealed Air Corporation. The bulk layers were produced similarly for all films used in the study, and only the external skin layer (5–10 μ m layer thickness) was modified. The commercial polymers used for this study include the polyethylene grade linear low-density (LLDPE) octene copolymer PE ExxonMobil Exact 3024, SK Primacor 1410 EAA, BASF Ultramid C33 PA6/66 polyamide copolymer, and the Kuraray EVAL E171B, a 44 mol% grade EVOH.

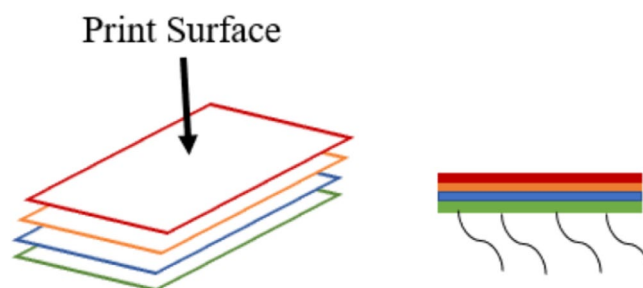


FIGURE 2 | Illustration of multilayer films and exterior printing surface of interest.

2.1.2 | Printing

The films were printed using a water-based polyurethane ink system prepared by Sealed Air Corporation. The ink consists of the following components: pigment, polymer, co-solvent, humectant, surfactant, and water, with an approximate total solids content of less than 20wt% and an initial surface tension of less than ~50 dynes per centimeter upon application. The printing involves a piezoelectric four-color CMYK process. The ink and the printing process are described in US Patent Applications US20230118001A1 [21], US 2024/0327662A1 [22], US 2024/0010019A1 [23], and US Patent US11008478B2 [24]. The films were corona-treated in line with a treatment setting of 1 W/ft² (10.76 W/m²) where applicable. Films were printed on the exterior layer only. All bulk layers remained the same throughout the study, and only the surface layer was modified, as illustrated in Figure 2.

2.1.3 | Contact Angle

SFE was evaluated using contact angle measurements collected using a Krüss MSA One-Click SFE surface analyzer. Sessile drop experiments were conducted using water (deionized UHPLC water, Sigma Aldrich, CAS 7732-18-5) as the polar probe liquid and diiodomethane (ReagentPlus, 99%, Sigma Aldrich, CAS 75-11-6) as the dispersive probe liquid, with a 2 μ L drop volume. The Krüss ADVANCE software was used to estimate the contact angle values from the captured images using the Young-Laplace fitting method. The surface free energies for the films were calculated using the OWRK (Owens, Wendt, Rabel, and Kaelble) method [25].

2.1.4 | Ink Adhesion and Durability

Ink durability testing was completed immediately after printing using the ASTM F2252-03 tape adhesion test method [26]. A piece of $\frac{3}{4}$ " 3M #600 clear pressure-sensitive tape was applied across the entire printed surface. The tape was then pulled with a steady, smooth motion at a 45° angle. The tape was then assessed for ink transfer, and a score was assigned based on the amount of ink present on the tape. Rub testing was conducted using a Testing Machines Inc. Digital Ink Rub Tester 10-20 using ASTM D5264 [27], where samples are mounted to a sled with the printed side in contact with Kraft paper. The unit then oscillates the sled back and forth for the desired cycle time. For this study, the oscillation cycles were 2 min long, with 100 cycles/min for 200 cycles. The characteristic adhesion and durability tests selected for this study are based on the commercial usage conditions, specifically printed ink resistance to scuffing due to repeated external stressors, such as rubbing contact with other materials during packaging and transport. In this case, the rub testing method provides a representative evaluation of the ink's contact durability in packaging applications.

2.1.5 | Atomic Force Microscopy (AFM)

The surface topographies and phase contrast imaging of the film samples were investigated using a Dimension Icon AFM (Bruker). A silicon NCHV probe with 42 N/m of spring constant and 320 kHz of frequency was used in all experiments, and the film sample was mounted on glass slides with double-sided tape. AFM height and phase scans ($5 \times 5 \mu\text{m}^2$ and $50 \times 50 \mu\text{m}^2$) were collected using tapping mode under ambient air conditions.

2.1.6 | Fourier Transform Infrared Spectroscopy (FTIR)

The samples were characterized using FTIR in attenuated total reflectance (ATR) mode to examine the functional groups present within the film surface. The equipment was a Nicolet iS20 FTIR spectrometer (Thermo Scientific, USA) equipped with a SMART iTX ATR accessory (Thermo Scientific, USA) and a diamond crystal. All spectra were analyzed using OMNIC (Thermo Scientific, USA) software. Each sample was run with 32 scans, with a spectral range of $1500\text{--}600 \text{ cm}^{-1}$ and a spectral resolution of 0.25 cm^{-1} .

2.1.7 | X-Ray Photoelectron Spectroscopy (XPS)

XPS analysis was performed using a Physical Electronics PHI VersaProbe 5000 station XPS to obtain survey and high-resolution (HRES) spectra of the neat and corona-treated polymer films. The XPS instrument is equipped with a monochromated 25.0 W Al 1486.6 eV source. To assess elemental composition, survey spectra were collected at a pass energy of 117.4 eV and a resolution of $\pm 0.5 \text{ eV}$. For all elements observed in the survey scan, HRES spectra were collected at a pass energy of 23.5 eV and a resolution of $\pm 0.2 \text{ eV}$. All XPS data were analyzed

using CasaXPS software (version 2.3.23PR1.0). Atomic percentages were determined from the survey spectra after processing each elemental region with Shirley backgrounds and applying the Physical Electronics instrument relative sensitivity factors (RSF), which are instrument dependent. All HRES spectra were charge-corrected by pinning the C1s aliphatic (C–C/C–H) band to 285 eV and were processed with Shirley backgrounds and CasaXPS's LA line shapes, which combine Gaussian and Lorentzian peak shape contributions that are typical for polymers [28].

3 | Results and Discussion

3.1 | Determining Surface Characteristics Before and After Corona Treatment

3.1.1 | FTIR

As the printing process depends on the surface characteristics, the film samples were characterized by FTIR and contact angle measurements (and subsequent calculations of SFE) before and after corona treatment. FTIR spectra for all untreated polymer samples are shown in Figure 3. For neat PE (control), peaks at ~ 2850 and 2920 cm^{-1} are attributed to C–H stretching, and peaks at 1460 and 720 cm^{-1} are associated with C–H bending and rocking modes, respectively [29, 30]. For EAA, the peak at $\sim 1700 \text{ cm}^{-1}$ is the identifying peak for this material, which corresponds to the carbonyl stretch of carboxylic acid [29, 30]. For PA, the peaks of interest are 1635 and 1537 cm^{-1} , which correspond to N–H bending in amines [28–31]. EVOH is a copolymer of PE and polyvinyl alcohol (PVA). A strong peak at 3307 cm^{-1} indicates –OH stretching, which is characteristic of PVA in the EVOH copolymer [30, 31]. Additionally, a peak at 1430 cm^{-1} is associated with the C–H bending mode in PE in the EVOH copolymer [30, 31].

Corona treatment directs a high-frequency discharge at the polymer surface, leading to surface oxidation and the generation of polar functional groups such as carbonyl groups, among others previously mentioned [10, 13, 32]. These generated polar groups lead to higher surface energy compared to untreated PE and improve the likelihood of adhesion between the liquid and the polymer film. The presence of polar groups such as hydroxyls or carbonyls is desired at the surface and can contribute to increased ink durability performance.

FTIR characterization was conducted after corona treatment for each sample type (Figure 3). Distinct changes in the FTIR spectral peaks due to the corona treatment process were expected. Still, they were not distinctly discernible in this case, as no additional peaks or shifts in peaks were detected (see Figure S1, ESI). The corona treatment settings used in this study were a quick-treat, or in-line corona treatment unit on the printing press. Correspondingly, the specimens were exposed to corona treatment for a short time ($< 1 \text{ s}$). Due to the short treatment times, the corona treatment was expected to lead to chemical functionalization over a very thin surface layer. FTIR measurements cannot capture the surface layer exclusively, and the responses from the bulk are dominant in the FTIR spectra obtained here. Instances of marked changes in FTIR spectra are reported in the

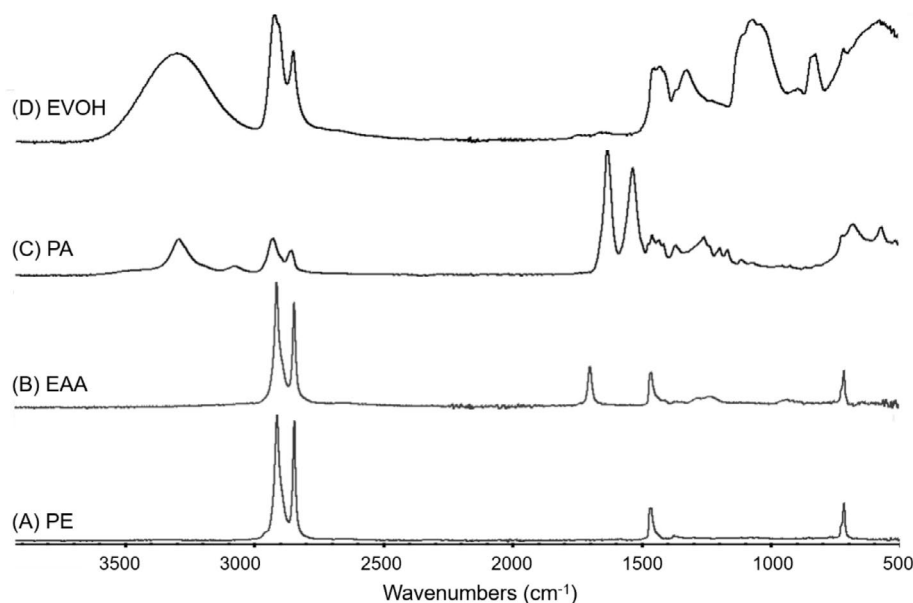


FIGURE 3 | FTIR spectra of untreated polymer films PE (A), EAA (B), PA (C), EVOH (D).

TABLE 1 | Surface free energy (SFE) of polymer substrates before and after corona treatment.

Polymer substrate	Calculated SFE (dynes/cm) before corona treatment	Calculated SFE (dynes/cm) after corona treatment
PE	29.1 ± 1.4	48.8 ± 1.7
EAA	33.7 ± 0.6	42.2 ± 1.8
PA	50.9 ± 2.1	58.4 ± 1.9
EVOH	43.4 ± 1.2	50.1 ± 2.3

literature where polymer films were subjected to corona treatment for an extended period of time, with some studies reporting treatment times of seconds—as low as 5 s [33]—and others up to 50 h [34], which are not representative of industrial in-line corona treatment processes (Figure 3). Studies with treatment times similar to this study (~1 s) have also been reported in the literature, where treated samples did not display any significant changes in spectral patterns [12].

3.1.2 | Contact Angle

As FTIR could not capture the change in the chemical composition at the surface, contact angles before and after corona treatment were performed, and the effect of corona treatment on SFE was determined. After measuring the contact angles, Young's equation and the Owens, Wendt, Rabel, and Kaelble (OWRK) method, where water was the polar probe liquid and diiodomethane was the dispersive probe liquid, were used to calculate SFE [25].

Table 1 summarizes the SFE values for the pre- and post-corona treated films. As expected, SFE values for the PE and EAA samples were in the range of untreated polyethylene (28–33 dynes/

cm) [10, 13] before corona treatment and increased to ~42 dynes/cm after corona treatment. PA and EVOH, both of which have naturally higher surface energy compared to PE or EAA, exhibited an increase in SFE with corona treatment, but the increase was not as large as was observed for PE and EAA. All samples responded as expected to the corona treatment and demonstrated SFE values in the expected ranges for the untreated and treated films. While FTIR spectral analysis was unable to determine the changes in surface chemical functionality, the SFE changes due to corona treatment and incorporation of polar functional groups were demonstrated through the increase in calculated SFE via contact angle data. AFM and XPS characterization studies were also conducted to understand the effects of corona treatment further. These results are discussed below.

3.1.3 | XPS

Neat and corona-treated polymer films were analyzed by XPS. Both survey and high-resolution (HRES) spectra were collected to assess the atomic composition and chemical bonding changes in the polymer films due to the corona treatment. Comparison of the HRES spectra for C 1s and O 1s regions, as well as the ratio of O 1s/C 1s atomic percentages, indicates a change in chemical bonding and an increase in oxygen content for the PE, EVOH, and PA films, while changes in the monolayer EAA film are ambiguous.

As summarized in Table 2, the oxygen content increased after the samples were corona-treated, which was expected. For example, the PE oxygen content of the PE sample after corona treatment was 18.25%, compared with 1.49% before corona treatment. Also, the carbon content decreased from 98.50% before the corona treatment to 74.47% after the corona treatment. This is expected due to the known functional groups introduced by corona treatment (C=O, —OH, —OOH, and —COOH) [10, 13, 32], which explains the increase in the oxygen content as the pre-treatment surface contained only CH₂ functional groups.

The high-resolution scans (HRES) were also investigated and are detailed in Figures S2–S7, and Tables S1–S4 ESI. For PE, the C1s and O 1s HRES spectra indicated an increase in peak intensities for CO (eV) and C=O/COOH (eV) functional groups, and a broadening of the overall C1s peak was observed after corona treatment. The EAA sample C1s and O1s HRES spectra for the corona-treated film showed lower contributions of acrylic acid functional groups. The C1s HRES peaks in the neat film are slightly lower than expected for EAA and PAA-based polymers in the literature. For EVOH, the presence of a carbonyl peak at 288.9 eV in the C1s HRES and a peak at 533 eV in the O1s spectra after corona treatment suggests the formation of carboxylic acid groups. In the PA films, the HRES N1s spectra for the corona-treated film show a significantly weaker polyamide peak at

~400 eV compared to the untreated film. An increase in the area percentages for peaks in the ~385.5–386.5 eV region in the C1s HRES spectra associated with C–O/C–OH/O–C–O suggests an increase in oxygen species on the corona-treated polyamide surface. The survey scan spectra for the samples are shown in Figure 4, and selected HRES spectra are shown in Figure 5.

3.2 | Surface Morphological Analysis

AFM was used to examine the changes in surface topography of the film samples before and after corona treatment and how the surface topography relates to the ink adhesion behavior of the polymer films. AFM height and phase images are shown in

TABLE 2 | Atomic compositions (%) for the polymer films before and after corona treatment as determined by XPS.

Element	PE		EAA		EVOH		PA	
	Untreated	Treated	Untreated	Treated	Untreated	Treated	Untreated	Treated
O 1s	1.49	18.25	1.43	3.66	11.28	26.58	5.93	10.64
C 1s	98.51	74.47	97.61	94.13	87.94	72.72	87.43	85.76
N 1s	—	—	—	—	—	0.31	4.52	—
O/C ratio	0.02	0.25	0.01	0.04	0.13	0.37	0.07	0.12

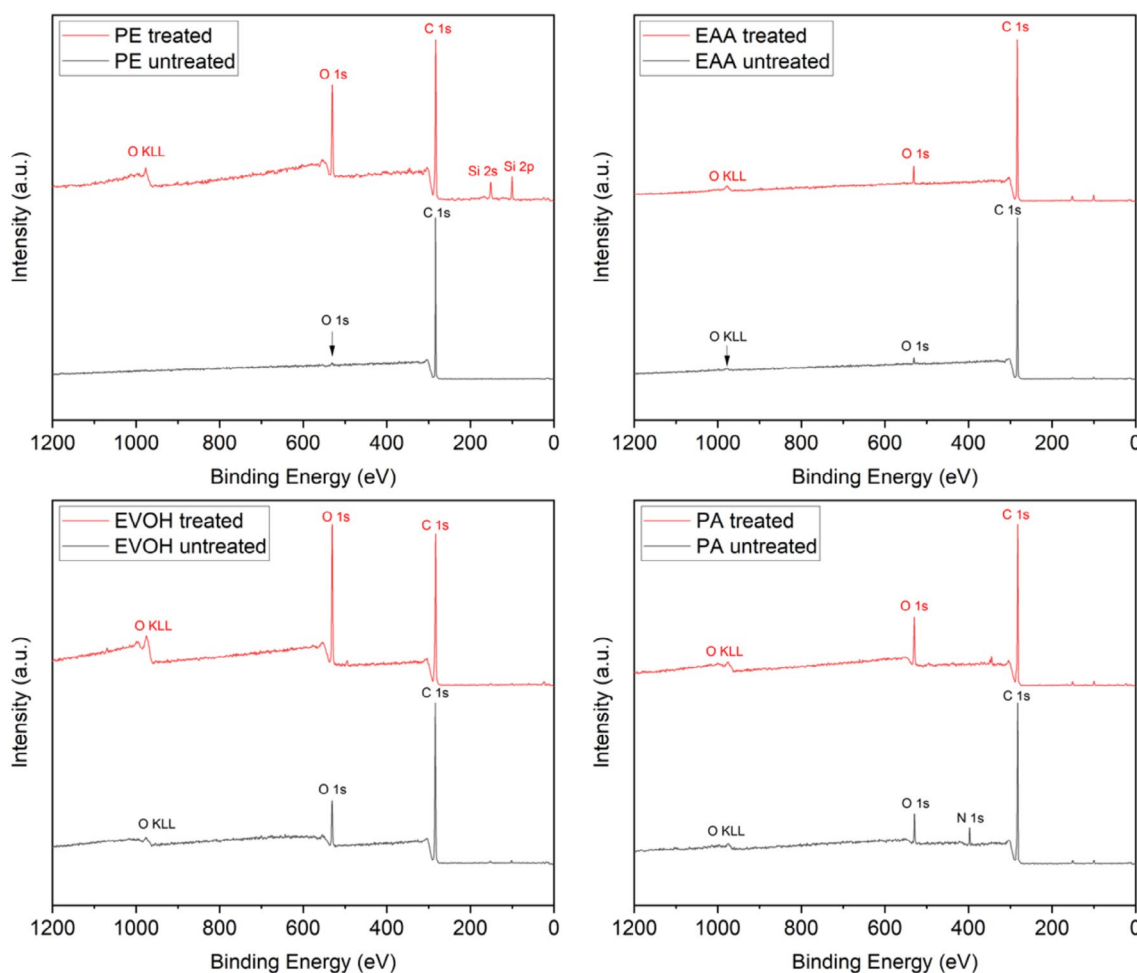


FIGURE 4 | XPS survey spectra of untreated and corona-treated polymer film samples.

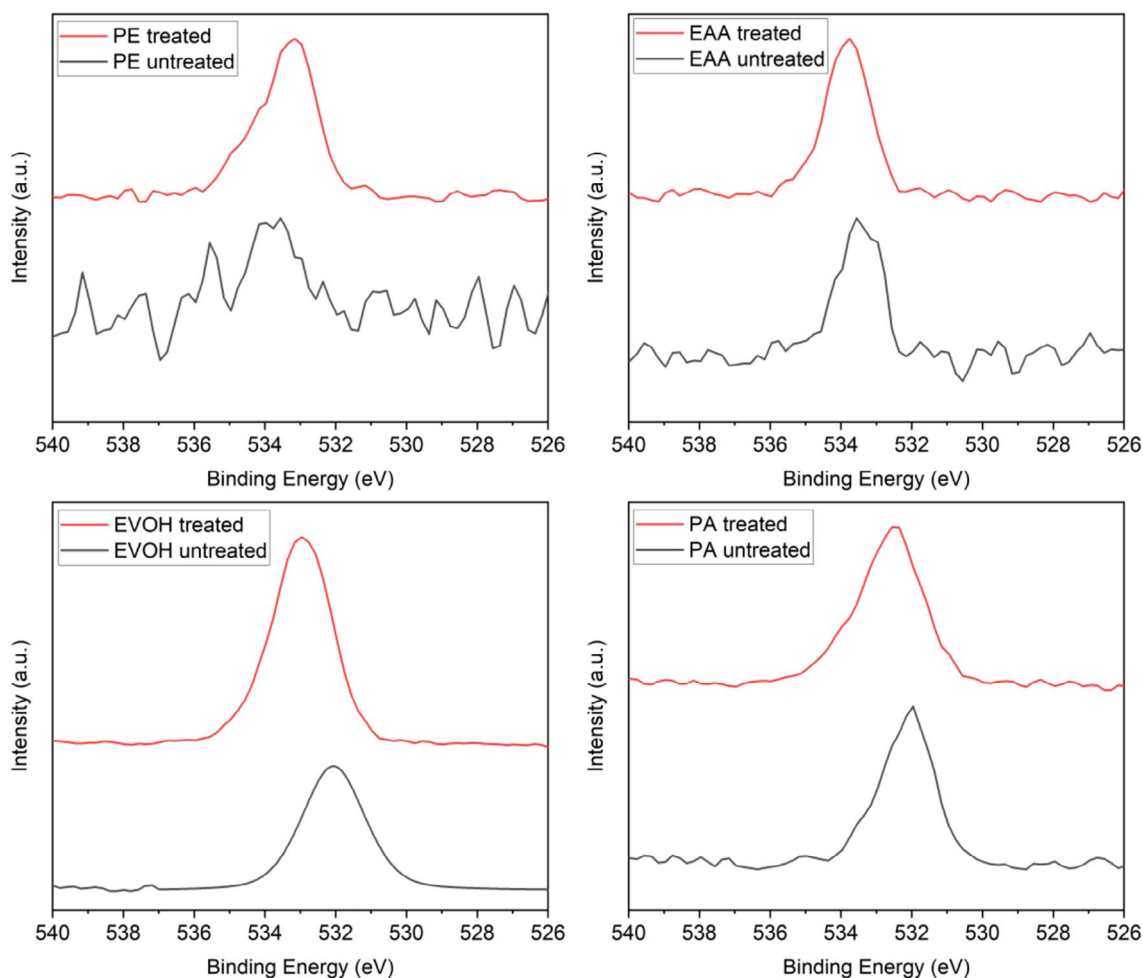


FIGURE 5 | HRES spectra for O1s regions of untreated (black) and corona-treated (red) films.

Figures 6 and 7, respectively. Untreated samples, specifically PE, display fibrous features, which have also been reported previously in the literature [35–39]. This has been attributed to crystallites present in PE, affected by the orientation of the polymer chains during processing and subsequent quenching. After corona treatment, the PE, EAA, and EVOH films exhibit a distinct change in surface topography, where the surface of the film appears to have globule-like structures, or “globular mounds”. These were not present before treatment but have been reported in the literature for polypropylene after corona treatment [35–39]. These mounds have been hypothesized to be low molecular weight oxidized material (LMWOM) [35–39] and have also been found to be water-soluble through several characterization methods. The similarities in appearance between the results presented by Strobel et al. [35] for PP and the samples in this study are notable. Given the similarities in structures between the PE and PP, the presence of LMWOM on the surface of the samples in this study could be assumed; however, with water-based ink, the treatment effects would be expected to contradict an improvement in ink adhesion.

The PA sample shows very minimal variation between untreated and treated surfaces. Literature data suggest PA should have been impacted more by corona treatment [33, 40]. This has been shown by XPS analysis with increased oxygen-containing functional groups [33]. Peng et al. [40] reported AFM images

with the appearance of surface mounds similar to the aforementioned results on PP, PE, EAA, and EVOH, but with corona-treated PA. In our study, while the increase in oxygen-containing functional groups was observed in the XPS analysis, the change in surface topography was not as pronounced as in the other three polymer-type films. The corona treatment process used in this study is an in-line process within the printing press, and the film is treated as it passes through the press, which is the most practical method of treatment for films in preparation for industrial printing. In the case of PA, the film may have needed a longer exposure to corona treatment to have an adequate effect to demonstrate observable changes in film surface morphology and chemistry. This is beyond the scope of the work of this study.

3.3 | Ink Adhesion and Durability

The films were printed in-line using a digital printing press and evaluated immediately for ink adhesion and durability using the tape-test method and rub evaluation as described in the methods section. An example of the printed image before testing and examples of tape test results are shown in Figure 8.

The ASTM tape standard test defines both slow and fast tape pulls to determine ink adhesion on a printed surface [26]. The test method is described in more detail in the materials and

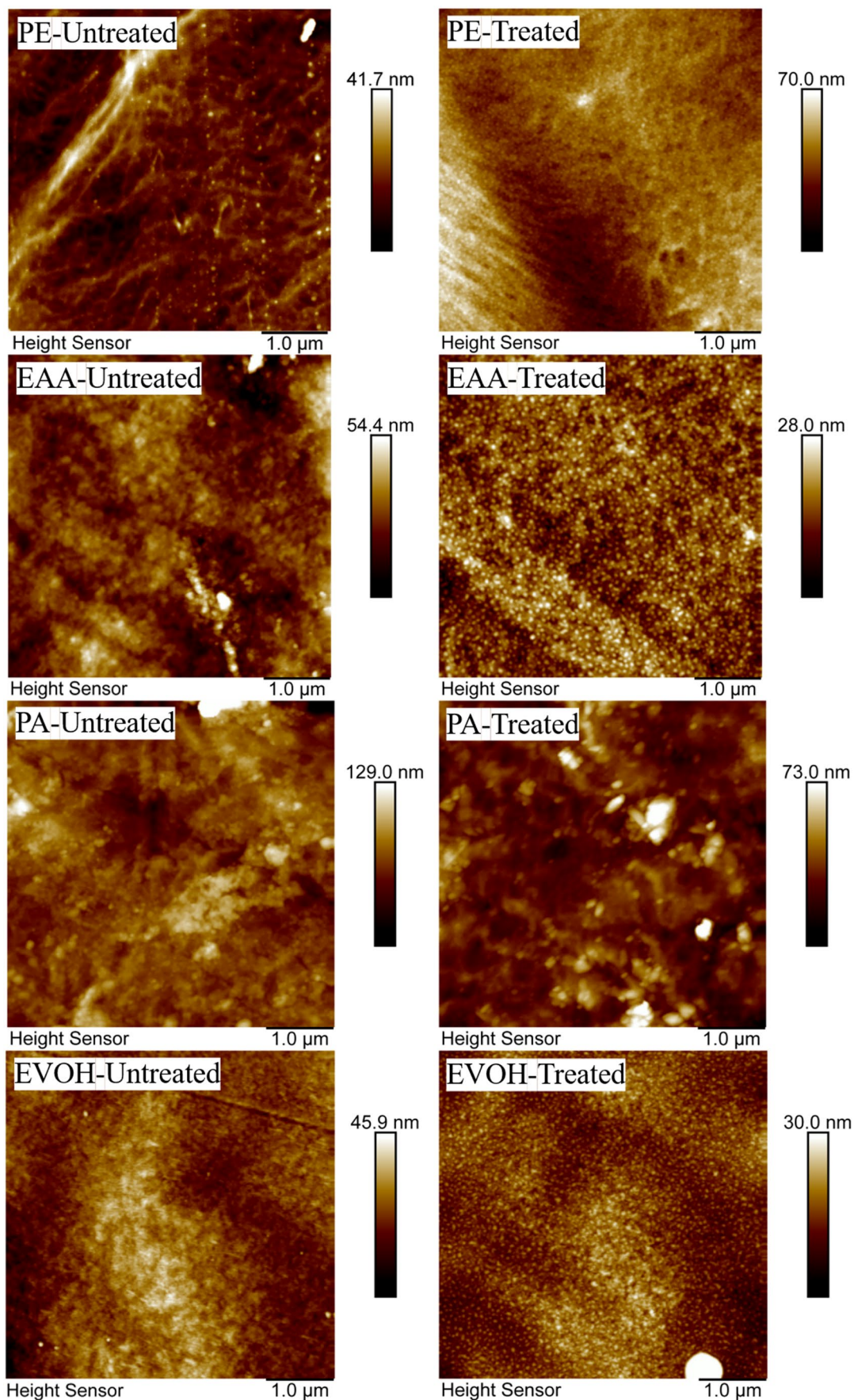


FIGURE 6 | AFM height scans (5 × 5 μm) showing the topography of untreated (left column) and treated (right column) PE, EAA, PA, and EVOH film samples.

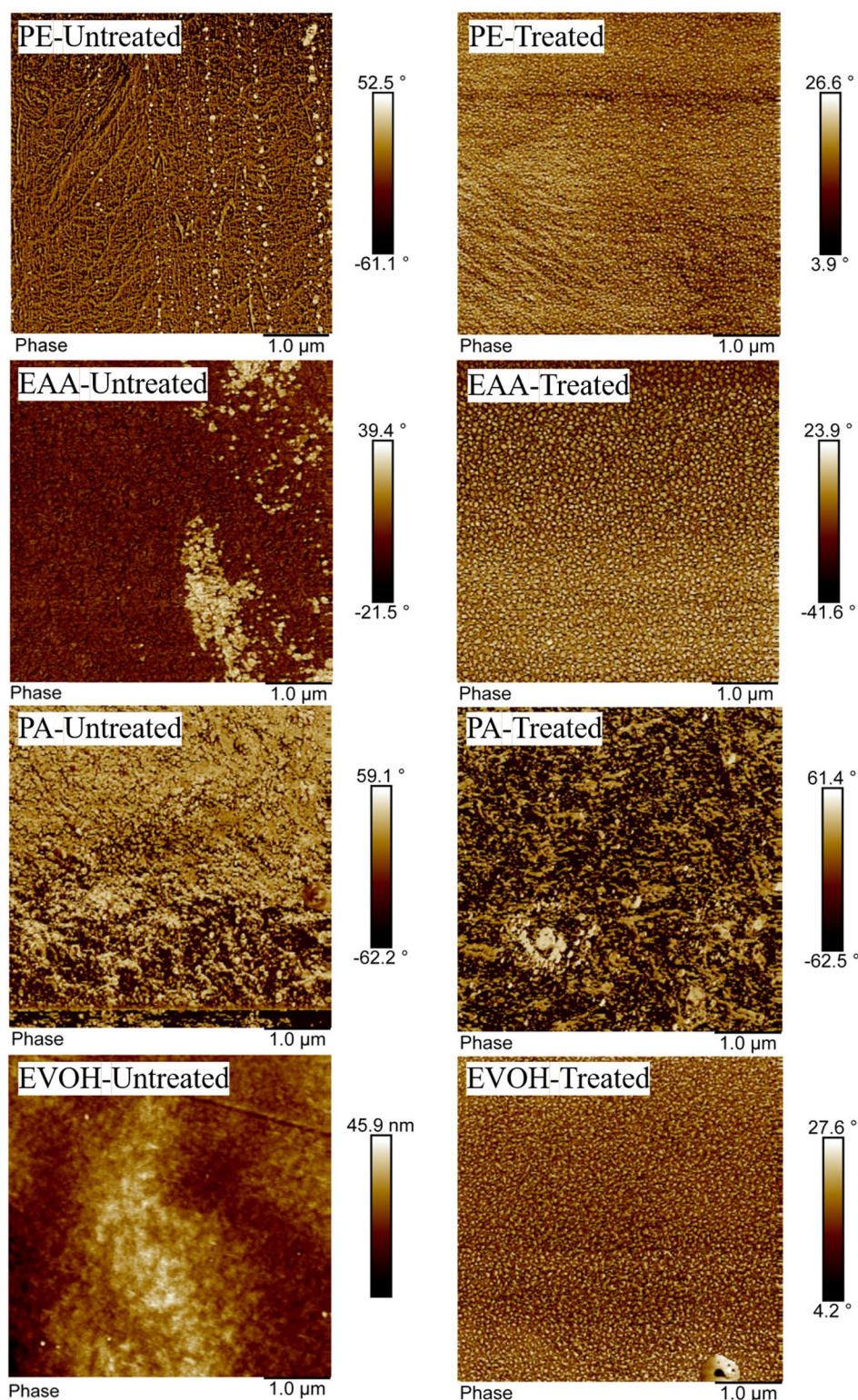


FIGURE 7 | AFM phase images of untreated (left column) and treated (right column) film samples.

methods section. In summary, after an image is printed onto a surface (Figure 8A), a piece of tape is applied to the printed surface smoothly and then pulled off either slowly or quickly according to which test is being conducted. This test method is a measure to assess the strength of adhesion between the ink and the print surface. The amount of ink that is on the tape after it is removed is an indication of the ink's adhesion to the surface. Figure 8B–E are images of the tape after slow and fast tape pulls. If there is a

high percentage of ink on the tape and little ink left on the film, as shown on the untreated images in Figure 8 (left side; untreated PE [B] and untreated PA [D]), the ink adhesion is poor. An indication of good ink adhesion would result in little to no ink on the tape after it is pulled off the printed area, as seen in Figure 8C, where the tape test results of treated PE are depicted. Tape test results of treated PA are shown in Figure 8E. After the tapes were pulled, the amount of ink on the tape was evaluated and scored according

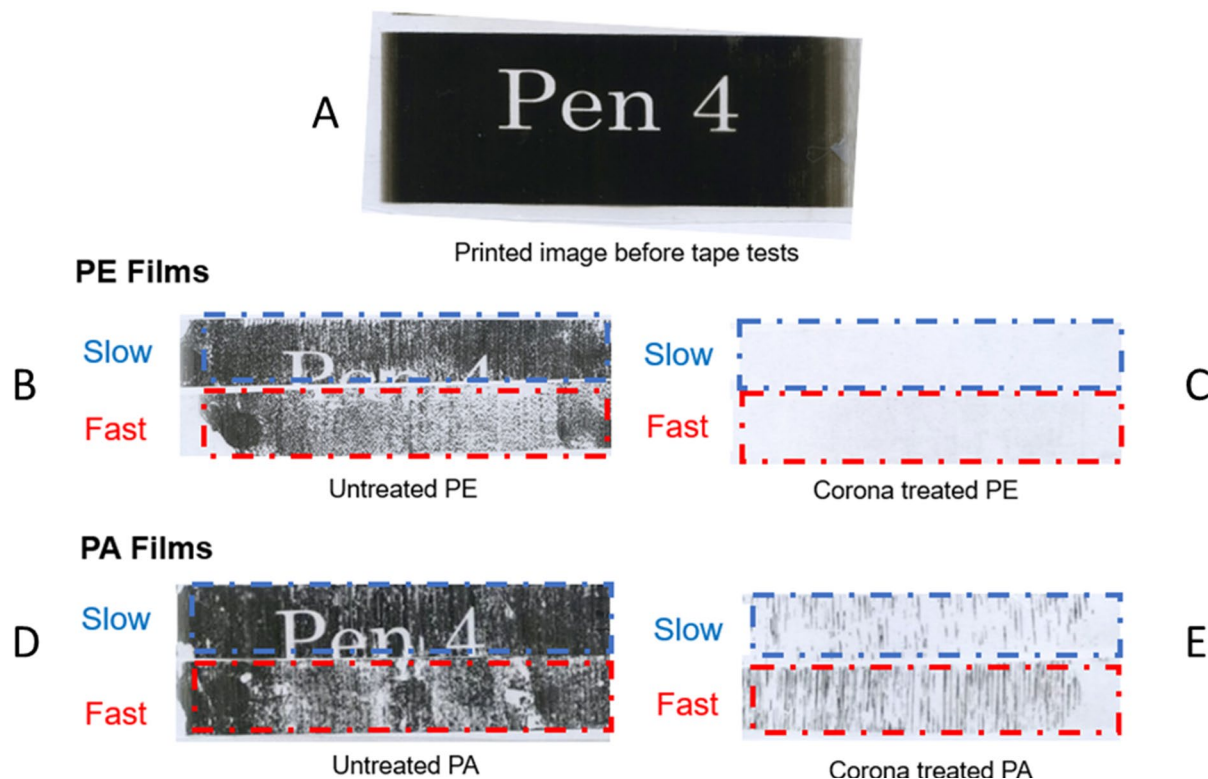


FIGURE 8 | Printed image graphic before adhesion and durability testing (A) and ink adhesion tape test results (B–D). The bottom panel of the figure illustrates the results of slow and fast tape tests of representative untreated and treated films.

TABLE 3 | Durability rating scale and tabulated ink adhesion and durability results by polymer type and corona watt density treatment level.

Durability rating scale	Substrate	Corona (W/ft ²)	Slow tape	Fast tape	Rub
Ink test scale (1–5)	PE	0	1	1	3
		1	5	5	4
5 = 0% ink loss	EAA	0	1	1	3
4 = 1%–25% ink loss		1	4	4	4
3 = 25%–50% ink loss	PA	0	1	1	3
		1	3	3	3
2 = 50%–75% ink loss	EVOH	0	1	1	1
1 = 75%–100% ink loss		1	4.5	5	1

Here, green signifies an acceptable score for the tape or rub test. Yellow signifies a marginal score of the tape or rub test. Red signifies an unacceptable score and very poor adhesion or durability for the tape or rub test.

to the Durability Rating Scale shown in Table 3. A summary of the results of tape testing and rub evaluations by polymer type and corona treatment level is also available in Table 3.

As shown in Table 3, there were varied results between the polymer films with respect to ink adhesion and durability. If only

the SFE were used as guidance, all corona-treated film samples would be expected to outperform untreated PE and EAA. Further, it was anticipated that the higher native SFE of the PA and EVOH film samples would demonstrate improved performance in untreated film ink adhesion compared to the untreated polyolefin films. However, our results indicate that the corona-treated PE resulted in the best ink adhesion scores, while the corona-treated PA resulted in the poorest post-treatment ink adhesion scores.

The ink adhesion and durability testing results are as anticipated for the treated PE and EAA samples. As a result of corona treatment, both PE and EAA indicated a marked increase in SFE in addition to observed topographical changes through AFM. The adhesion and durability results support the guideline practiced in industry.

However, the adhesion and durability results for the PA and EVOH samples are contradictory to the hypothesis, where films with surface energies in the range of corona-treated PE would exhibit better ink adhesion and durability even without corona treatment. The PA and EVOH films both demonstrated improved ink adhesion with corona treatment in the tape test, but did not demonstrate durability improvement in rub testing scores, even after corona treatment. The PA samples were poorer in durability scores after corona treatment than all other samples. As observed in the contact angle/SFE, AFM, and XPS characterizations, the PA film exhibited minimal changes pre- and post-corona treatment. The surface topography was relatively unaffected, as observed in the AFM images (Figures 6 and 7). The SFE was minimally affected by corona treatment in the PA films, and while the XPS analysis

did indicate a small increase in oxygen content, the increase was not as significant as the better-performing (with respect to ink adhesion and durability) PE and EAA films. EAA showed a small increase in oxygen content as analyzed by XPS (an increase of 2.23% after treatment) but demonstrated a more pronounced increase in SFE as determined by contact angle evaluation. EVOH showed a more pronounced increase in oxygen with XPS analysis, with an increase of 15.32% in oxygen content post-treatment.

The results above suggest that the oxygen content, and more importantly, the reactive groups present on the surface along with surface topography, are more influential for ink adhesion and durability than factors such as high native SFE. While it can be observed that all four polymer types studied were affected by the corona treatment, it can be justifiably concluded that polymer types have varied responses to corona treatments, resulting in varied surface topographies, surface chemistries, and ink adhesion and durability capacity. The types of functional groups, surface density of the functional groups, and reactivity of the functional groups, combined with surface topography, directly influence the level of ink adhesion and durability of the polymer surfaces, signifying that the degree of ink adhesion and durability is more complex than SFE guidelines used in industry. While PE was the most promising substrate for printing in this study, future investigations would be necessary to understand the effects of comonomer content, molecular weight, branching, and other physical attributes of PE grades.

4 | Conclusions

Industry guidelines for acceptable ink adhesion and durability currently reference SFE differentials as the primary factor in predicting successful printability. Based on this guideline, the SFE of a print substrate should be in the range of 40–50 dynes/cm to generate successful ink adhesion with a representative water-based ink. The initial hypothesis of this work was that polymer films having native SFE values that are in the same range as those achieved by corona treatment of polyethylene, ca. 41–46 dynes/cm, will have significantly higher ink adhesion and rub-resistance durability compared to a standard polyethylene, with typical SFE values of 28–33 dynes/cm. However, the findings from this study show that polymers with native SFE in the range of 41–46 dynes/cm, due to their chemical composition and surface functional groups, did not demonstrate improved ink adhesion or durability. All polymers tested required corona treatment to demonstrate any notable improvement in ink adhesion and durability. The analysis methods, including ATR-FTIR, contact angle goniometry with resulting SFE calculations, AFM, XPS, and ink adhesion and durability testing, established baseline information for untreated samples and demonstrated surface functionality changes that were expected in the corona-treated samples, confirming the surfaces were adequately treated and prepared for printing. Combining the results of ink adhesion and durability testing with other analytical data, it was determined that the influence of corona treatment through oxidation, production of surface-active functional groups present at the surface, and topographical modifications are more significant factors for predicting acceptable

ink adhesion and durability than the SFE differences between the ink and the polymer substrates. Polymers with high native SFE values in the range of corona-treated PE (ca. 41–46 dynes/cm) (i.e., PA, EVOH) did not demonstrate better or improved ink adhesion or durability versus polymers with lower native SFE values (i.e., PE, EAA, ca. 28–33 dynes/cm) either pre- or post-corona treatment. The observed changes in corona-treated PE and EAA, including oxidation and changes in surface topography, in combination with a higher SFE, led to higher ink adhesion and durability performance in comparison to PA and EVOH. From these results, it can be concluded that surface chemistry and surface topography combined are more influential factors than SFE alone as a means of achieving improved ink adhesion and durability. These observations and corresponding conclusions provide updated guidance for flexible packaging converters, as the industry's rule of thumb for using ink and substrate SFE differentials to predict acceptable ink adhesion is not adequate.

Author Contributions

Ashley C. Wynne: conceptualization, investigation, writing – original draft, methodology, validation, writing – review and editing, visualization, formal analysis, data curation, resources, funding acquisition. **Morgan A. Smith:** investigation, resources. **Kayla Foley:** investigation, methodology, writing – review and editing, formal analysis, data curation. **I-Wei Chu:** methodology, investigation. **Miles A. Johnston:** investigation. **Keisha B. Walters:** investigation, writing – review and editing, supervision. **Santanu Kundu:** conceptualization, funding acquisition, writing – original draft, methodology, visualization, writing – review and editing, software, project administration, supervision, resources.

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Data Availability Statement

Additional data supporting the findings of this study will be made available upon request from the corresponding author.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** pen70167-sup-0001-supinfo.docx.